IMPERIAL
SCHOOL OF PUBLIC HEALTH

Digital Media Platform: An Intervention Implementation Proposal

CID: 06031009 – *Addressing Opioid Misuse Among Veterans in Rural Central-Appalachia*
18th December 2024

Foundations of Public Health Practice *Project Title: Pathways to Opioid Relief: Behavioural Interventions to Address Opioid Misuse Among U.S. Veterans in Rural Central-Appalachia*

Contents

1 Context & Introduction.....1

2 Aim & Objectives.....2

3 Conceptual Justification.....3

4 Key Platform Features.....4

4.1 Work Packages..... 5

4.2 Project Timeline..... 6

4.3 Stakeholder Engagement.....7

4.4 Resource Planning & Budgeting.....8

5 Process Evaluation.....9

6 Risk Management.....10

6.1 Residual Risks & Monitoring.....11

7 Population Health Value & Next Steps.....12

The opioid crisis in the Central-Appalachian Region is severe and wicked, especially in rural areas.

Population Health Challenge: Problem & Context

- In 2021, 1 in every 22 deaths in the U.S. were attributable to unintentional opioid-toxicity, highlighting the persistent societal burden of the opioid epidemic (1)
- Constituting ~6% of the country's adult population, the crisis has disproportionately affected U.S. veterans, who face psychological, physiological, and socio-structural vulnerabilities stemming from barriers to accessing care (2-5)
- Systematic review by the U.S. Department of Veterans Affairs (VA) highlights that veterans are twice as likely to die from opioid overdose compared to the general population (6)
- Member states of Central-Appalachia are particularly affected *(Appendix A),* with rural areas experiencing a disproportionate burden due to socio-economic vulnerabilities and limited healthcare infrastructure (7,8)
- Use of prescription and illicit opioids to manage chronic pain stemming from service-related injuries drives this crisis among veterans, emphasizing the need for targeted interventions
- While effective for use against pain, prolonged use carries significant risks of dependence, misuse, and eventual overdose (9)
- Geographical barriers, therapy provider shortages, and limited telehealth infrastructure inhibit the reach of national and state-wide initiatives in rural Central-Appalachia, including the Opioid Safety Initiative (OSI) and state-level Prescription Drug Monitoring Programs (PDMPs) (8,9-11)

Targets: Population & Behaviour

This wicked population health problem therefore accentuates the need for interventions which address underlying behavioural drivers among **veterans in rural Central-Appalachia** contributing to misuse behaviours while **promoting adoption of evidence-based non-opioid pain-management strategies.**

We propose *'Veteran Compass'*, an integrated digital media platform, as one possible solution.

Overall Project Aim

To promote the adoption of non-opioid pain-management strategies among U.S. veterans in the target population, reducing reliance on opioids and addressing related health disparities

Project Objectives relating to *Veteran Compass*

1. To re-identify relevant behavioural barriers, change components, intervention functions, and policy categories to promote the positive target behaviour in the target population
2. To propose *Veteran Compass*, a digital media platform intervention that incorporates behavioural insight to facilitate positive behavioural change
3. To outline work packages, timeline, management, and basic governance in proposed intervention implementation
4. To establish *Veteran Compass* in partnership with key stakeholders for veteran-use in rural Central-Appalachia to promote the target behaviour in the target population, improving population health outcomes
5. To serve as a blueprint for implementation in other areas of the United States

The Platform incorporates behavioural insights to reduce region-wide veteran opioid-reliance.

Behavioural Analysis: COM-B & the Theoretical Domains Framework

Behavioural analysis identified barriers in capability, physical (*C-Ph* (skills) and psychological (*C-Ps* (knowledge) barriers faced by veterans living in rural Central-Appalachia among others through the COM-B and Theoretical Domains (TDF) frameworks

- It was found that physical disability and presence of comorbidity can negatively affect reflective motivational *(M-Re)* beliefs about self-efficacy around accessing care (8,12)
- This is exacerbated by a lack of awareness of alternative care options and VA benefits among rural veterans, impacting access of care among those with and without severe disability (12)
- Poor digital literacy, (in some) coupled with misconceptions around efficacies of alternative strategies, further limits engagement with existing telehealth services and reinforces a preference for opioid-use for pain-management (12)
- Geographic isolation from specialized care facilities, unreliable transportation, and socio-economic factors that contribute to systemic poverty and misuse behaviour in rural Central-Appalachia only worsen these issues

Behavioural Analysis: The Behavioural Change Wheel

Barriers identified through COM-B and TDF were systematically mapped to nine possible intervention functions to identify which functions could contribute the most positive change toward the target-behaviour:

- Education – to address awareness knowledge, misconceptions, and health literacy
- Training – to improve digital literacy and technology skills
- Enablement – to facilitate access to telehealth among isolated veterans and those with severe disability

Communication/Marketing and Service Provision policy categories were found to be best suited to utilize the identified intervention functions after assessment with APEASE criteria to enable effective delivery

- Use of these categories contributed greatly to the success of state-level interventions against veteran opioid-misuse in Kentucky and West-Virginia, two of the most concerning member states of Central-Appalachia (Appendix A) (13,14)
- *Veteran Compass* will utilize these intervention functions and policy categories to deliver positive population health outcomes

4.0 Key Platform Features

The platform employs a nudge-based approach to address underlying behavioural determinants.

Accessibility

- Compatible with phones, tablets, & computers
- Downloadable content for offline use
- Intuitive & visually appealing user-interface (UI) for veterans with varying technological familiarity

Behavioural Nudges

- Default reminders & progress feedback to encourage healthy habits
- Social validation through peer recognition & highlighting of community milestones & abstinence success stories
- Daily exercises to create habitual routines for pain-management without opioids

Personalization

- Customizable user profiles & learning paths tailored according to individual needs based on self-report & user activity
- “Near Me” feature to identify and display care centres nearest to user location
- Allerts & reminders for follow-up appointments, learning goals, and daily, weekly, or monthly health tips

Education & Awareness *(C-Ps)*

- Interactive modules will break down non-opioid pain-management strategies into palatable and digestible tutorials
- Health literacy content will inform veterans about opioid risks, alternative therapies, and available care options such as telehealth, cognitive behavioural therapy (CBT), physical therapy, and medications like methadone and buprenorphine and their efficacies (15)
- Gamified Learning will utilize points, online badges, and activity streak leaderboards to incentivize user engagement and persistency
- A Feedback System will provide reminders and personalized suggestions to reinforce and improve learning

Skills Development & Tutorials *(C-Ph)*

- Digital literacy training will be provided through tutorials regarding access and use of underutilized regional telehealth services, troubleshooting devices, and access of online resources
- Video conference guidance will educate veterans on navigation through video conference tools, particularly VA services
- Pain-management tools will include tutorials on physical therapy and exercises

Community & Support

- Forum and Chat Functions will facilitate peer discussions enabling sharing of successful veteran stories, experiences, and encouragement, potentially providing modelling and boosting optimism
- Moderated Spaces will employ trained facilitators to guide conversations and provide support while bypassing geographical barriers, travel costs, and time constraints

4.1 Work Packages

The platform will employ a four-pronged approach to implementation in rural Central-Appalachia.

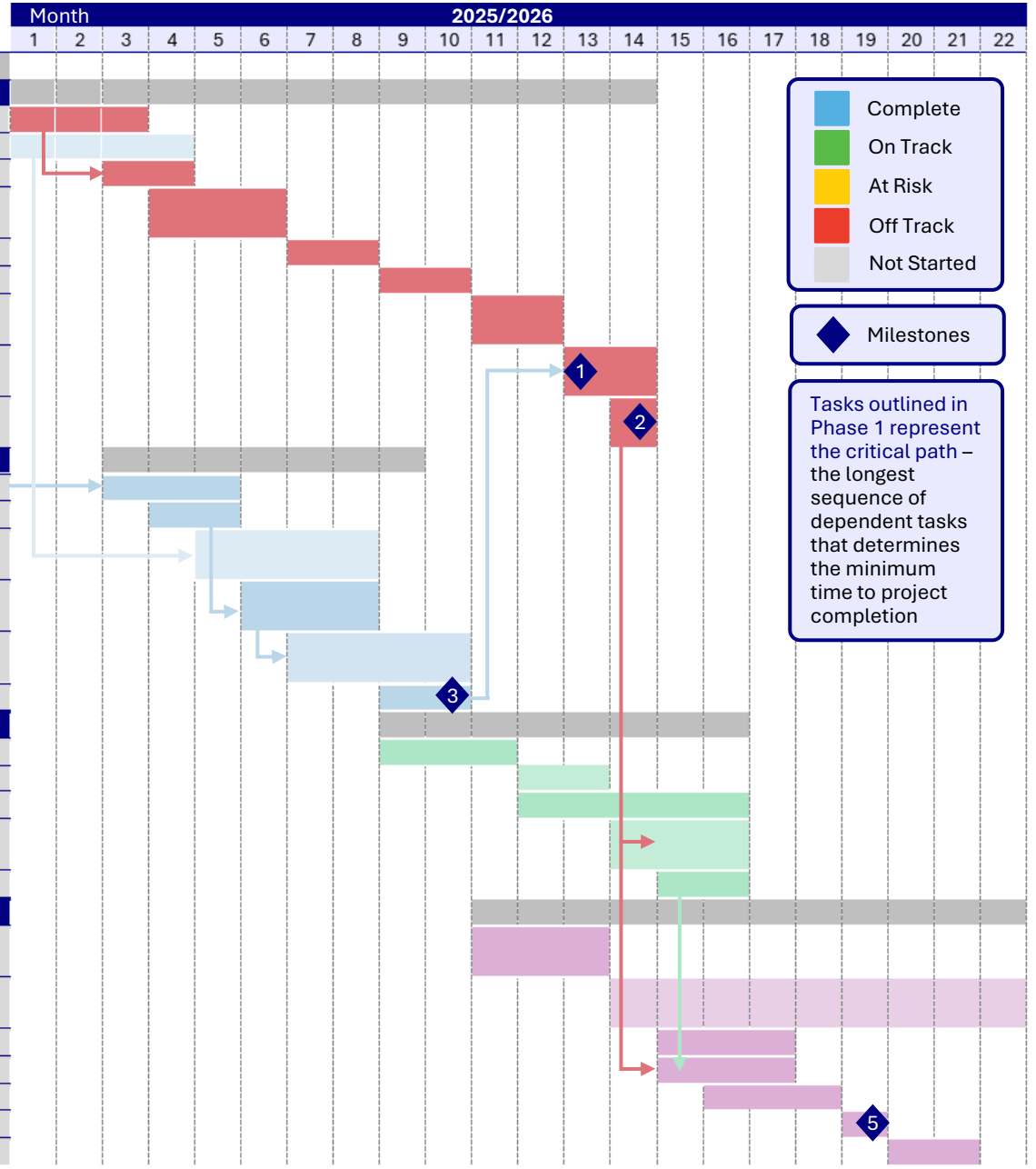
	1 Design & Development	2 Content Creation	3 Outreach & Training	4 Monitoring & Evaluation
Activities	Define front-end (UI) and back-end (database and servers) platform infrastructure, incorporating identified key features and ensuring scalability Create an intuitive and attractive design accommodative of varying levels of digital literacy among veterans, prioritizing user testing to refine navigation, accessibility, and mobile responsiveness	Collaborate with VA and specialized healthcare professionals, behavioural psychologists, and veterans successful in adoption of non-opioid pain management strategies to design engaging educational content utilizing infographics, videos, and interactive quizzes Research and adapt resources addressing digital literacy, telehealth navigation, and care centre locations	Launch marketing and outreach campaigns through collaboration with veteran-focused networks, VA (pension, disability compensation, and heath service programs) initiatives, commissaries, and community organizations to encourage initial engagement. Webinars and live demonstrations would build further confidence and interest Host online training sessions on platform use, basic digital literacy, and telehealth services focused on foundational skills such as accessing modules, participating in forums, and troubleshooting	Use analytics to track Veteran Compass usage patterns, such as churn and retention rates, monthly/daily active users (MAU/DAU), quiz completion rates, and forum activity. Regularly Compare pre and post quiz metrics to assess module effectiveness Identify barriers to engagement through user feedback and adapt platform or outreach strategies accordingly
Deliverables	Interractive and attractive Veteran Compass platform prototype	Comprehensive initial content library	User registration counts and conversion rates, identification of referral sources, and click-rates of outreach strategies	Monthly progress reports and process evaluation reports at end of intervention implementation
Milestones <Milestone no.>	Beta-testing <1> and platform completion <2>	Content library <3> ready to support education, training, and outreach	150-200 veterans signed-up <4> (reflecting the typically small percentage of initial intervention adopters)	Publication of evaluation report <5> assessing what's been done, and what needs to be
Dependencies	None; it serves as foundation for all subsequent activities	Requires general but not necessarily complete definition of software infrastructure	Dependent on platform design & development, and completion of content and beta version	Requires sufficient user-data acquisition through outreach and marketing success

4.2 Project Timeline

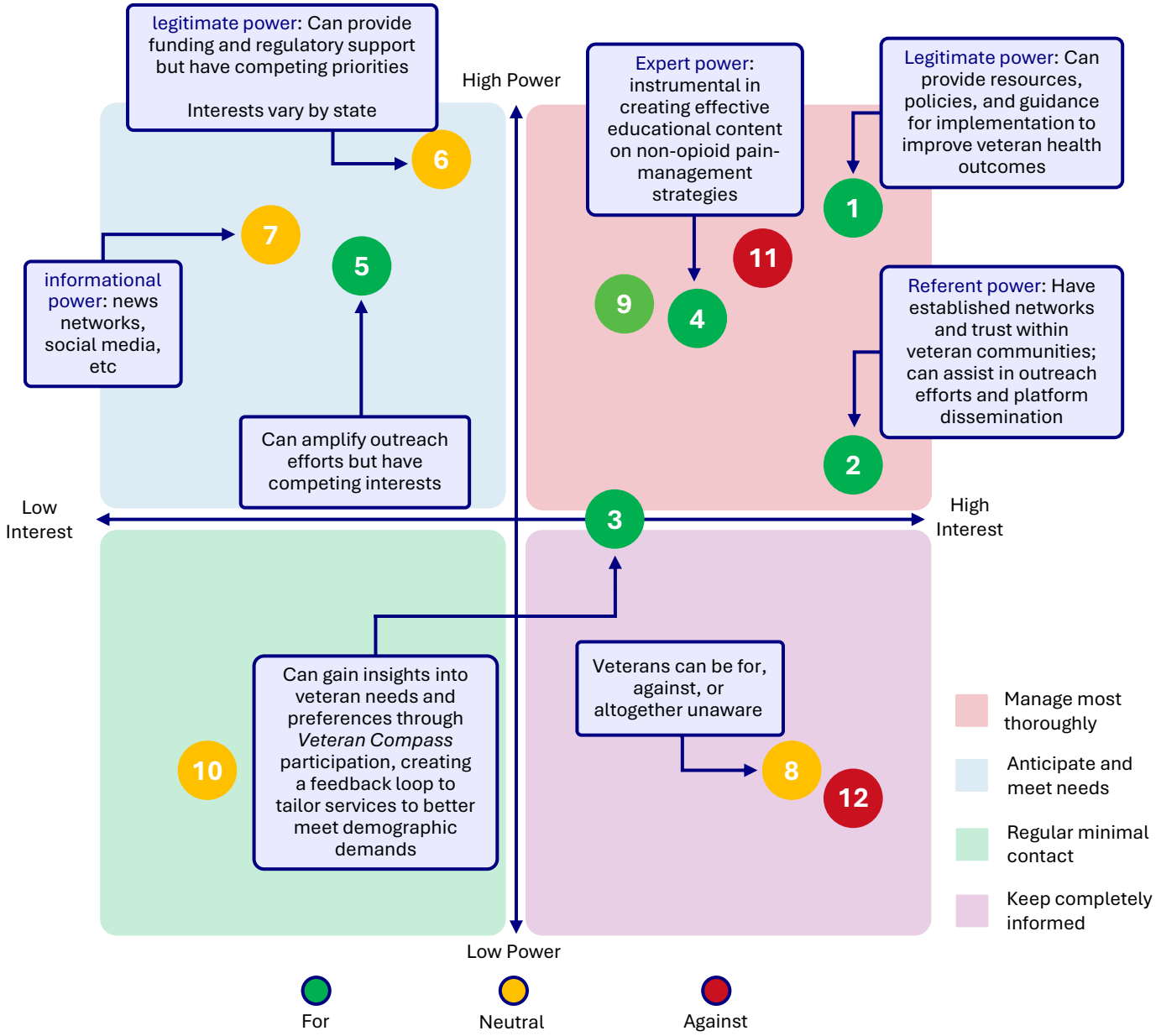
Implementation & process evaluation is expected to span roughly 2 years.

Senior Project Leader (PL) (Democratic Leader)

	Activities	Assignment	Start	End	Status
	Phase 1: Development & Design				
1.1	Securing Funding	Grant Writer & Project Manager (PM)	01/01/2025	31/03/2025	
1.2	Stakeholder Engagement	PM & Outreach Lead	01/01/2025	30/04/2025	
1.3	Recruitment of UI/UX Developers	Human Resources (HR) Lead	01/03/2025	30/04/2025	
1.4	Defining front & Back-End Infrastructure	Software Designers, UI Developer(s), Software Engineers, Database Specialists	01/04/2025	30/06/2025	
1.5	Prototype Creation	UI/UX Developers	01/07/2025	31/08/2025	
1.6	Conduction of User-Testing	Testing Co-ordinators & Veteran Focus Groups	01/09/2025	31/10/2025	
1.7	Platform Refinement for Accessibility & Responsiveness	UI/UX Developers	01/11/2025	31/12/2025	
1.8	Beta Testing	PM, Testing Co-ordinators, & Veteran Focus Groups	01/10/2026	28/02/2026	
1.9	Veteran Compass Platform Completion & Release	Entire Development Team	01/02/2026	28/02/2026	
	Phase 2: Content Creation				
2.1	Ethical Approval for Content	PM & Ethics Co-ordinator	01/03/2025	31/05/2025	
2.2	Recruitment of Content Creation Specialists	HR Lead	01/04/2025	31/05/2025	
2.3	Collaboration with VA & Powerful Experts for Content Development	VA, Content Specialists, Behavioural Psychologists, & Veterans	01/05/2025	31/08/2025	
2.4	Infographics, Videos, & Interractive Quizzes Design & Development	Content Specialists, Software Developers, Graphic Designers, Videographers	01/06/2025	31/08/2025	
2.5	Research and Adaptation of Digital Literacy Resources	Research Team	01/07/2025	31/10/2025	
2.6	Completement of Full Content Library	Content Specialists & Digital Archivist	01/09/2025	31/10/2025	
	Phase 3: Marketing & Training				
3.1	Developing Marketing Strategies	Marketing Lead & Marketing/Outreach Team	01/09/2025	30/11/2025	
3.2	Recruiting Platform-Training Facilitators	HR Lead	01/12/2025	31/01/2026	
3.3	Launching & Running Marketing Campaigns	Marketing Team & Veterans Networks	01/12/2025	30/04/2026	
3.4	Conducting Online Digital Literacy Training Sessions	Outreach Team & Training Facilitators	01/02/2026	30/04/2026	
3.5	Hosting Webinars & Live Demonstrations	Training Facilitators	01/03/2026	30/04/2026	
	Phase 4: Monitoring & Evaluation				
4.1	Selecting, Integrating, & Defining Analytic Tools & Metrics	Data Analyst	01/11/2025	31/01/2026	
4.2	Tracking Platform Usage Metrics and Monthly Reports	PM & Data Analyst	01/02/2026	Ongoing	
4.3	Comparing Pre & Post-Quiz Metrics	Data Analyst	01/03/2026	31/05/2026	
4.4	Collecting & Assessing User Feedback	Feedback Co-ordinator	01/03/2026	31/05/2026	
4.5	Process Evaluation	Evaluation Team	01/04/2026	30/06/2026	
4.6	Stakeholder Meeting for Evaluation Report	PM & Data Analyst	01/07/2026	31/07/2026	
4.7	Platform Refinement According to Insight	Development Team	01/08/2026	30/09/2026	



Key stakeholders will be considered throughout implementation & beyond.



Stakeholders

1. Veterans Health Administration (VHA) and VA
2. Veteran Service Organization (VSOs)
3. Telehealth Service Providers
4. Behavioural Psychologists and Healthcare Professionals
5. Community Organizations in Appalachia
6. Federal and State Government Agencies and Policy Makers
7. Regional Media
8. U.S. Veterans in Rural Central-Appalachia
9. Local Healthcare Providers
10. General rural Central-Appalachian Public
11. Opioid-dispensing pharmaceuticals
12. Illicit opioid-dealers

- There are more stakeholders than has been highlighted; the above are just a few
- Stakeholders have varying degrees of different types of power and interest
- Stakeholder positions are not stagnant

Funding requirements based on U.S. averages estimate costs at roughly $2 million.

Operational: $167k

- Office Rent & Utilities: $45k (over 15 months)
- Software Licensing & Subscriptions: $15k (design tools, project management, & analytics over 15 months)
- Miscellaneous Equipment: $7k (printing, training materials, etc.)
- Promotional Campaigns: $50k (social media advertisements, posters, flyers, & engagement campaigns)
- Outreach & Marketing Travel: $15k
- User Testing & Focus Groups: $15k (facilitation, materials, & veteran travel)
- Contingency Reserve: $20k

Staff: $1.476Mil

- Grant Writer: $16.5k
- Senior PL: $315k
- PM: $96k
- HR Lead: $36k
- Software Designers: $42k
- Software Engineers: $54k
- Software Developers: $105k
- Database Specialists: $48k
- Development Team: $70k
- Outreach Staff: $57k
- Marketing Staff: $112.5k
- Ethics Co-ordinator: $19.5k
- Focus-Group Co-ordinators: $54k
- Testing Co-ordinators: $52k
- Training facilitators: $49.5k
- Data Analysts: $112k
- Process Evaluation: $33k
- Other Staff: $204k

Core: $57k

- Ethical Review Fee: ~$5k (application & processing)
- Stakeholder Engagement Meetings: $12k (organization, venues, potential travel)
- Report Preparation, Documentation, and Dissemination: $40k (mid-phase and final reporting)

Capital: $205k

- *Veteran Compass* Platform Development Infrastructure: $150k (including scalable serve setup & initial software/hardware investments)
- Logistical Technology: $30k (laptops, monitors, & peripherals for staff & team)
- Contingency Reserve: $25k

USD $1,905,000 in Implementation Costs

These costs do not include maintenance and ongoing costs; they are costs related to *Veteran Compass* intervention implementation activities

See (Appendix B) for staff-cost calculations

5.0 Process Evaluation

We can evaluate *Veteran Compass* implementation through the LogFrame.

	Summary	Indicators	Means of Verification	Risks & Assumptions
Goal	To reduce opioid-use for pain-management among U.S. veterans in rural Central-Appalachia	- Regional opioid dispensing rates, indicative of prescription-opioid use - Overdose deaths, indicative of illicit-opioid use	- Centers for Disease Control (CDC) opioid dispensing rate maps - National Center for Health Statistics (Death Certificates)	- Assumes opioids being dispensed are being used by veterans - Assumes reported related deaths are representative of illicit-use (which they are not)
Purpose	**To promote the adoption of non-opioid pain-management strategies among U.S. veterans in the rural Central-Appalachians** while addressing key underlying behavioural predictors to opioid-misuse	- Increased engagement with non-opioid strategies and services among veterans across the region	- Partnerships/collaboration with key stakeholders such as VA, VHA, VSO, and telehealth service providers	- Assumes relevant stakeholders will participate
Outputs	- Provide an easy-to-use, attractive, and social digital media platform utilizing a nudge-based approach - Address *C-Ph* barriers identified by behavioural analysis in the target population, utilizing education to improve health literacy - Facilitate *Veteran Compass* dissemination whilst accounting for *C-Ps* concerns (intimidation of digital tools due to poor digital literacy)	- % of users who return to app after first use over (retention rate) - Module and quiz completion rates - No. of veterans who signed up or marked interest in training sessions	- MAU/DAU engagement metrics and user feedback - Module and quiz completions logs and back-end app analytics - Proportion of veterans who follow through in attending webinars and training sessions	- Assumes applicability and effectiveness of nudge-based strategies across the entire rural Central-Appalachian region - Assumes sustained/effective educational-content engagement by veterans - Assumes veterans will engage with training sessions despite initial intimidation or lack of familiarity with digital media
Activities	- Creation of a defined *Veteran Compass* platform infrastructure with an interactive and attractive design - Creation of engaging and informative educational content incorporating adapted resources for educational content library - Successful launch of marketing and outreach campaigns alongside platform training sessions	- % of time platform is operational without system disruptions (system uptime and availability) - % of completed content items - % fraction of no. of app installs and sign-ups over total no. of users reached (conversion rate)	- Service level agreement (SLA) reports provided by hosting / cloud providers, and system monitoring tools - Content tracking and review logs, and eventual completion of comprehensive content library - Click rates and user registration counts outlined in marketing campaign reports	- Assumes allocated resources will be sufficient for defining infrastructure and effective in creating an attractive platform design - Assumes initial iteration of content library will cater to the entire region (generalizability) - Assumes indicators will reflect resonance in regional sub-populations

Mitigating key risks before they occur will be critical to sustained implementation success.

Budget Overruns

- Outlined as not likely but critical in impact through use of the risk register framework (Appendix C)
- **Risk:** Budget constraints relating to unanticipated costs or delays in activity phases
- **Mitigation:** Inclusion of contingency reserves and conductance of quarterly financial reviews to identify and address overspending

Content Development Delays

- Likely and moderate in impact
- **Risk:** Collaboration with various stakeholders can foreseeably lead to delays in content development and finalization
- **Mitigation:** Development of a phased content rollout plan beginning with core modules; overlapping tasks can facilitate development efficiency

Data Privacy & Security

- Likely but critical in impact
- **Risk:** Misuse/breach of sensitive veteran data
- **Mitigation:** Incorporation of stringent cybersecurity measures, encryption standards, regular security audits, and information governance to comply with HIPAA and other relevant regulations

Low Platform Adoption

- Expected and severe in impact
- **Risk:** Veterans facing stigma around opioid-misuse and mental health, alongside veterans intimidated by digital tools may be deterred to engage
- **Mitigation (i):** Launch of targeted awareness campaigns communicating community narratives and success stories from veterans who have benefited from non-opioid strategies
- **Mitigation (ii):** Simultaneous implementation of the ‘Peer Support Groups’ intervention across rural Central-Appalachia (as outlined in behavioural analysis) could address optimism, emotion, social influences, and beliefs about capabilities and consequences
- **Mitigation (iii):** Hosting webinars and live demonstrations will accommodate for veterans of varying digital literacy skills, introducing all to use of *Veteran Compass*

Rural Infrastructure Barriers

- Highly likely but moderate in impact
- **Risk:** Limited broadband access and technological infrastructure in more rural areas can hinder platform usability and engagement
- **Mitigation:** Integration of offline-capable features and partnerships with local ISPs and libraries

Good governance will ensure compliance with legal, regulatory, and ethical requirements.

Residual Risks, Monitoring, & Governance

- **Residual Risks:** Low adoption in certain counties may persist despite mitigation due to systematic distrust, cultural barriers, and limitations of promotional campaign reach. Digital infrastructure gaps present significant challenge to engagement in the most remote areas and communities of the rural Central-Appalachian region.
- **Monitoring Plan:** It has therefore been planned to track user metrics continuously, amass qualitative feedback through focus groups, and adjust outreach strategies and platform features based on insight.
- Stakeholder participation and consensus orientation, accountability, transparency, responsiveness, effectiveness, equity, and abidance to the rules of member State laws in risk management will ensure good governance, abating corruption and failure while enhancing accountability and transparency
- **Reputational Risks:** These will help mitigate against breaches in data privacy and security, ineffective or inaccurate content, interest misalignment, and target bias, which have potential to harm the trust, credibility, and public perception of *Veteran Compass* and stakeholders



Veteran Compass offers population health value, with potential for beyond.

Population Health Value: The Institute for Healthcare Improvement (IHI) Tripple Aim & Intervention Quality

- Ideally, *Veteran Compass* will improve opioid-use related health outcomes of the defined collective population, shaped by social, economic, and environmental determinants.
- It hopes to do this through an integrated platform, facilitating access to underutilized telehealth services while offering personalized, remote, consistent, and cost-effective care comparative to mobile clinics or contingency management across the vast region of rural Central-Appalachia.
- As such, it would enhance user experience and care-provider satisfaction in a manner that is timely, effective, person-centred, efficient, equitable, and safe.

Future Steps

- Outcome evaluation after a sufficient period of time post *Veteran Compass* implementation can emphasize practical significance and actionable insights through leveraged utility, propriety, and accuracy in accordance with stakeholder interests and cultural responsiveness.
- Impactability can be assessed to better understand the degree to which veterans in rural Central-Appalachian sub-populations (such as states, counties, etc.) will benefit from *Veteran Compass* via identification of those likely to respond positively to the intervention.
- Ongoing risk management, monitoring, and outcome and formative evaluations can then be used to identify strengths and weaknesses of the intervention in such sub-populations, assessing achievement of long-term goals and informing downstream strategies.
- If successful, transfer evaluations can be conducted to assess the replicability of *Veteran Compass'* mechanisms and outcomes for transfer into other notable regions of the U.S. facing similar challenges.

Veteran Compass could eventually evolve to become an integrated care platform, enabling secure sharing of medical information between different care providers, facilitating communication, reducing delays, and improving continuity of care to improve health outcomes and reduce inequalities in the veteran population.

References

1. Yeo Y, Johnson R, Heng C. The Public Health Approach to the Worsening Opioid Crisis in the United States Calls for Harm Reduction Strategies to Mitigate the Harm From Opioid Addiction and Overdose Deaths. Mil Med. 2022 Aug 25;187(9–10):244–7.
2. Schaeffer K. The changing face of America's veteran population. 2023 Nov.
3. Office of Research & Development; VA Research on Pain Management - Opioid Medications [Internet]. U.S. Department of Veterans Affairs. U.S. Department of Veterans Affairs; [cited 2024 Nov 25]. Available from: https://www.research.va.gov/topics/pain.cfm#research4
4. Edlund MJ, Steffick D, Hudson T, Harris KM, Sullivan M. Risk factors for clinically recognized opioid abuse and dependence among veterans using opioids for chronic non-cancer pain. Pain. 2007 Jun;129(3):355–62.
5. Bennett AS, Guarino H, Britton PC, O'Brien-Mazza D, Cook SH, Taveras F, et al. U.S. Military veterans and the opioid overdose crisis: a review of risk factors and prevention efforts. Ann Med. 2022 Dec;54(1):1826–38.
6. Use of Opioids in the Management of Chronic Pain Work Group, Department of Veterans Affairs, Department of Defense. VA/DoD Clinical Practice Guideline for the Use of Opioids in the Management of Chronic Pain. Washington, DC; 2022 May.
7. Tanz LJ, Gladden RM, Dinwiddie AT, Miller KD, Broz D, Spector E, et al. Routes of Drug Use Among Drug Overdose Deaths — United States, 2020–2022. MMWR Morb Mortal Wkly Rep. 2024 Feb 15;73(6):124–30.
8. Moody LN, Satterwhite E, Bickel WK. Substance use in rural Central Appalachia: Current status and treatment considerations. Journal of Rural Mental Health. 2017 Apr;41(2):123–35.
9. Why Health Care Is Harder to Access in Rural America. 2023 May.
10. Smith NZY, Thornton JD, Fenton SH, Simmons D, Champagne-Langabeer T. Helpful, Unnecessary, or Harmful: A Systematic Review of the Effects of Prescription Drug Monitoring Program Use on Opioid Prescriptions. Pharmacoepidemiology. 2023 Dec 15;2(4):350–65.
11. Andrea SB, Gilbert TA, Morasco BJ, Saha S, Carlson KF. Factors Related to Prescription Drug Monitoring Program Queries for Veterans Receiving Long-Term Opioid Therapy. Pain Medicine. 2021 Jul 25;22(7):1548–58.
12. Filteau MR, Green B, Jones K. Barriers to Community Treatment for Opioid Use Disorders among Rural Veterans. Journal of Veterans Studies. 2021 Nov 5;7(3):83.
13. Corey Davis N for PHL, Wesley Sargent C, Janice Vick I, Henrietta Kuoh C, Pierre-Olivier Cote C. Leveraging Prescription Drug Monitoring Program (PDMP) Data in Overdose Prevention and Response [Internet]. 2021 Mar [cited 2024 Dec 1]. Available from: https://www.cdc.gov/overdose-prevention/media/pdfs/2024/03/Leveraging-PDMPs-508.pdf
14. Wood N, Mells J, Dotson T, Jeffries JE. Enhancing and Leveraging the West Virginia's Prescription Drug Monitoring Program (PDMP) for Public Health Surveillance and Clinical Decision Making: A Case Study. Journal of Public Health Management and Practice. 2023 Mar;29(2):E37–43.
15. Mannes ZL, Stohl M, Fink DS, Olfson M, Keyes KM, Martins SS, et al. Non-pharmacological Treatment for Chronic Pain in US Veterans Treated Within the Veterans Health Administration: Implications for Expansion in US Healthcare Systems. J Gen Intern Med. 2022 Nov 19;37(15):3937–46.

IMPERIAL
SCHOOL OF PUBLIC HEALTH

Thank You

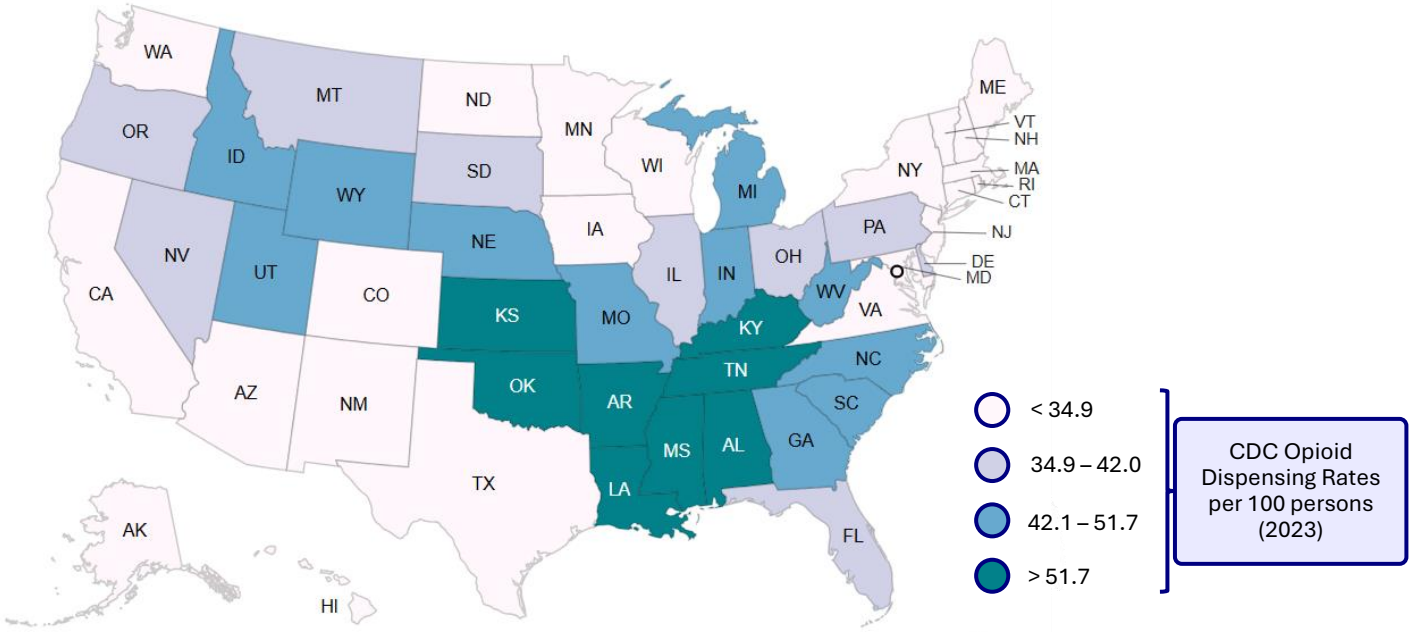
Any Questions?

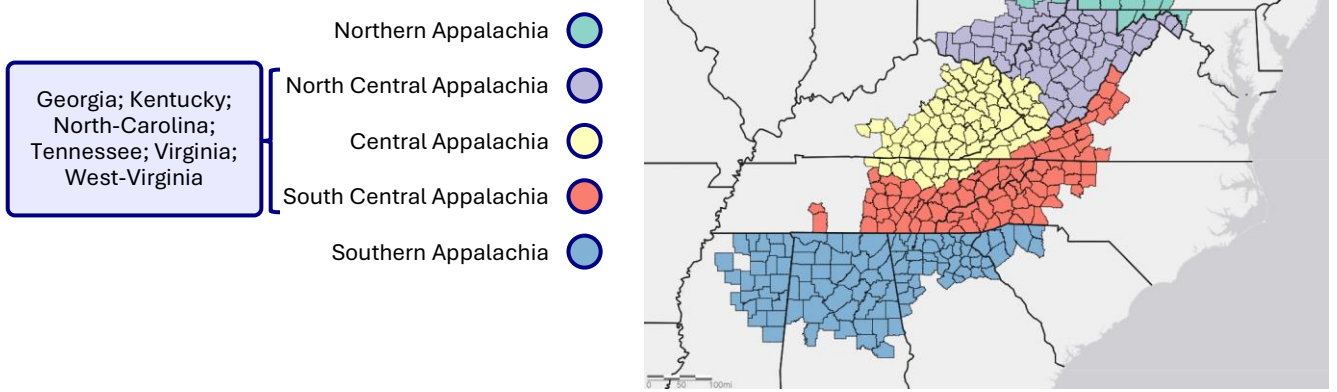
Veteran Compass Team
CID.06031009@veterancompass.us

We appreciate your continued support.

Appendix A

U.S. Opioid Dispensing Rates (2023) & Central-Appalachia





Appendix B Staffing Cost Estimations Breakdown

Project Phase	Staff Role	Monthly Estimate per individual	Quantity of Months Employed	Team Size	Total Months Employed
Phase 1	Gantt Writer	$5,500	3	1	3
	Project Manager (PM)	$8,000	6	1	6
	UI/UX (Software) Developers	$7,500	4	2	8
	Software Designers	$7,000	3	2	6
	Software Engineers	$9,000	3	2	6
	Human Resources (HR) Lead	$6,000	2	1	2
	Outreach/Communications Lead	$6,000	4	1	1
	Database Specialists	$8,000	3	2	6
	Testing Co-ordinators	$6,500	4	2	8
	Veteran Focus Group Co-ordinators	$4,500	4	3	12
	Development Team Members	$7,000	1	4	4
Phase 2	PM	$8,000	3	1	3
	Ethics Co-ordinator	$6,500	3	1	3
	HR Lead	$6,000	2	1	2
	Content Specialists	$5,500	5	3	15
	Graphic Designers	$6,000	3	2	6
	Videographers	$6,000	3	2	6
	Simple Research Team	$5,000	4	2	8
	Digital Archivist	$5,000	2	1	2
	Software Developers	$7,500	3	2	6
Phase 3	Marketing Lead	$7,500	3	1	3
	Outreach/Communications Team	$5,500	6	3	18
	HR Lead	$6,000	2	1	2
	Marketing Team	$6,000	5	3	15
	Video Conference Facilitators	$5,500	3	3	9
Phase 4	Development Team	$7,000	2	3	6
	Data Analysts	$8,000	7	2	14
	PM	$8,000	3	1	3
	Communications Co-ordinator	$5,500	3	2	6
	Public Health Process Evaluation Team	$5,500	3	2	6

i.e. Estimated Process Evaluation Staffing Costs = $5,500 × 3 × 2 = $33,000 USD

Appendix C The Risk Register

	Negligible	Minor	Moderate	Severe	Critical
Expected p > ~0.9					
Highly Likely p < ~0.9					
Likely p < ~0.6					
Not Likely p < ~0.3					
Slight p < ~0.1					

Where ‘p’ is the expected probability of occurrence

